Lecture 5: Forecast Uncertainty and Prediction Intervals

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Why point forecasts are not enough

Definition: Point forecast

A point forecast gives one best guess for the future, such as "next week's sales will be 510 units."

Definition: Prediction interval

A prediction interval gives a range of plausible future outcomes, such as "next week's sales will likely fall between 470 and 550 units."

Business decisions rarely depend only on the center of the forecast. A risk manager cares whether the lower tail of the return distribution might breach a drawdown limit, not just whether the average prediction looks reasonable. The tradeoff is precision versus honesty: point forecasts are simple to communicate, but intervals acknowledge that the future can vary around the center. For beginners, the key rule is that a forecast without uncertainty is incomplete.

Example 5.1. *The notebook plots daily S&P 500 returns over the test period together with the random forest's point forecast and 80%/95% prediction intervals. Days where the realized return falls outside the 80% interval are flagged, showing that the intervals are imperfect but substantially informative. The interval widens during high-VIX episodes because market uncertainty rises when volatility spikes.*

Residual-based interval construction

Definition: Residual uncertainty

Residual uncertainty is the unpredictable part of the forecast error that remains after the model uses all available information.

The simplest interval starts from the model's fitted forecast and adds an uncertainty band estimated from past forecast errors. This is appealing in practice because it uses a quantity students already know: residuals. The limitation is that the residual spread may not be constant across all periods. Still, residual-based intervals are a useful first step because they connect uncertainty directly to

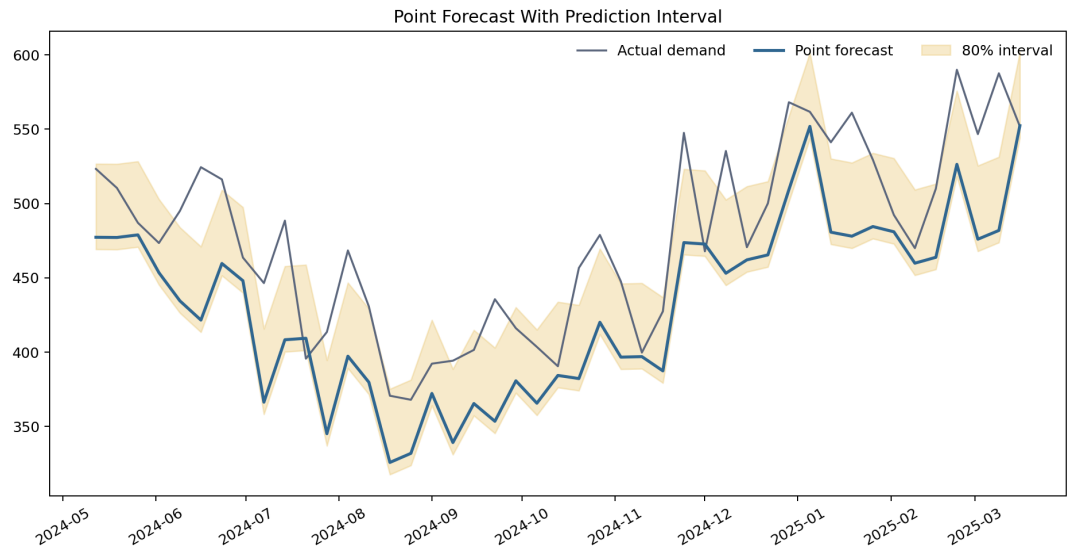

Figure 1: Example 5.1 notebook output: point forecast and 80%/95% intervals on S&P 500 daily returns.

forecast performance that students can observe.

$$\hat{y}_{t+1|t} \pm z_{\alpha/2}\hat{\sigma}_e$$

When to use this formula. This formula assumes forecast errors are approximately normally distributed. If residuals are skewed or heavy-tailed—common with promotions, outliers, or volatile financial data—the normal-theory z value can understate tail risk. A safer and more general approach is to use *empirical quantiles* of the out-of-sample residuals directly:

$$\text{Lower bound} = \hat{y}_{t+1|t} + Q_{\alpha/2}(\text{OOS residuals}), \qquad \text{Upper bound} = \hat{y}_{t+1|t} + Q_{1-\alpha/2}(\text{OOS residuals}),$$

where Q_p is the p-th quantile of the observed error distribution. The notebook uses empirical quantiles for this reason.

Example 5.2. *The notebook estimates interval bands from out-of-sample validation residuals rather than from in-sample fit. A histogram and Q-Q plot of these residuals reveal fat tails heavier than a normal distribution—a common property of financial return forecasting errors. This confirms that empirical quantiles are safer than the normal-theory z formula for equity return intervals.*

Coverage and sharpness

Definition: Coverage

Coverage is the fraction of realized outcomes that fall inside a prediction interval.

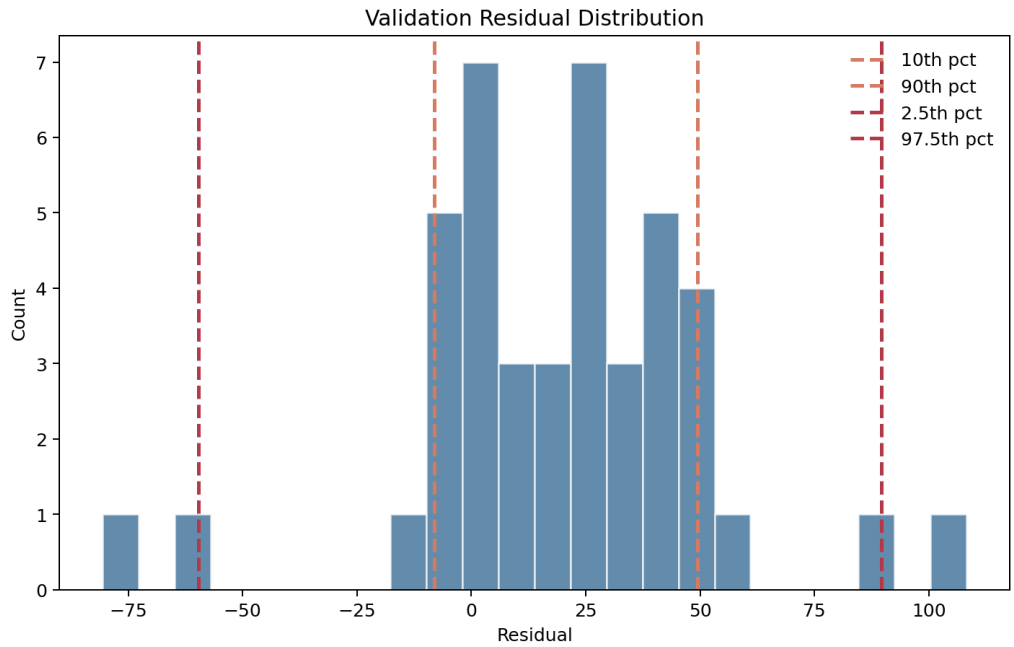

Figure 2: Example 5.2 notebook output: residual histogram and Q-Q plot showing fat tails in S&P 500 forecast errors.

Definition: Sharpness

Sharpness refers to how narrow the interval is. Narrower intervals are more informative, but only if coverage stays credible.

Good interval forecasts balance two goals. First, realized outcomes should fall inside the interval at roughly the advertised rate. Second, the interval should not be so wide that it becomes useless for decisions. The tradeoff is unavoidable: very wide intervals look safe but provide little action value, while overly narrow intervals sound confident but fail too often.

Example 5.3. *The notebook compares 50%, 80%, and 95% prediction intervals on the S&P 500 test set. Coverage rises as intervals widen, but the practitioner's question is whether the added width still provides actionable risk information: an interval spanning ±5% on a daily return forecast has limited value for position sizing.*

State-dependent uncertainty

Definition: Heteroskedasticity

Heteroskedasticity means forecast errors have different variance in different situations, such as calm weeks versus promotion weeks.

Uncertainty is often not constant. In equity markets, periods of low volatility are followed by

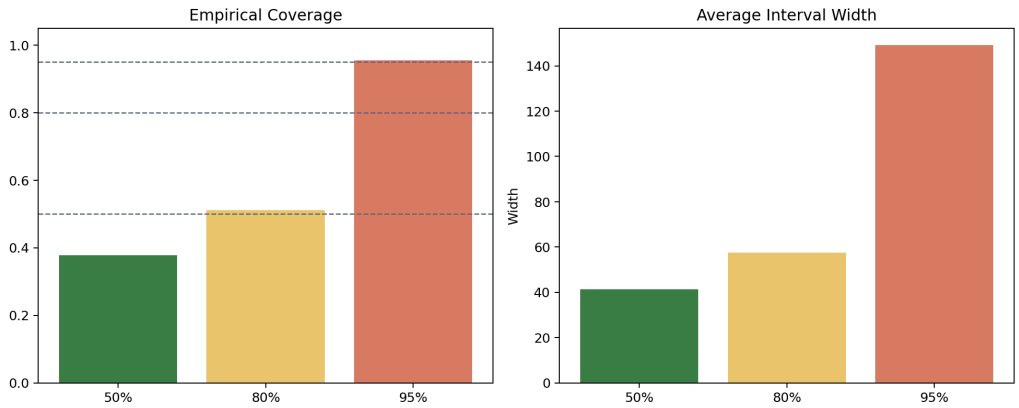

Figure 3: Example 5.3 notebook output: wider intervals improve coverage but reduce sharpness on S&P 500 returns.

sharp regime switches where daily return swings become much larger. For applied forecasting, this matters because one fixed interval width can systematically understate risk in volatile periods and overstate risk in stable periods. Students should interpret this as a timing problem: the same model can be "more uncertain" in some states of the world than in others.

Example 5.4. *In the notebook, validation residuals are grouped by VIX level: calm (VIX < 15), normal ($15 \leq$ VIX < 25), and stressed (VIX ≥ 25). Boxplots and interval-width comparisons show that stressed-market weeks produce forecast errors two to three times wider than calm weeks, which is direct evidence that uncertainty depends on market volatility conditions.*

Scenario forecasting

Definition: Scenario forecast

A scenario forecast asks how the predicted path changes when key assumptions change, such as price, promotion intensity, or weather conditions.

Scenario analysis is especially useful when the future depends on economic conditions that can move in different directions. Instead of pretending there is one fixed future, we compare a neutral market path, an optimistic bull scenario, and a downside bear scenario. The gain is decision relevance: a risk manager can ask "what if volatility spikes from here?" rather than relying on a single passive forecast. The cost is that scenario quality depends on the realism of the assumptions.

Example 5.5. *The notebook builds three 15-day forward scenarios starting from the last observation in the test set. The bull scenario assumes positive MSFT momentum and a declining VIX; the bear scenario assumes negative momentum and rising volatility; the neutral scenario holds all features at recent averages. Cumulative return paths and daily confidence bands for each scenario show how sensitive the forecast distribution is to market assumptions.*

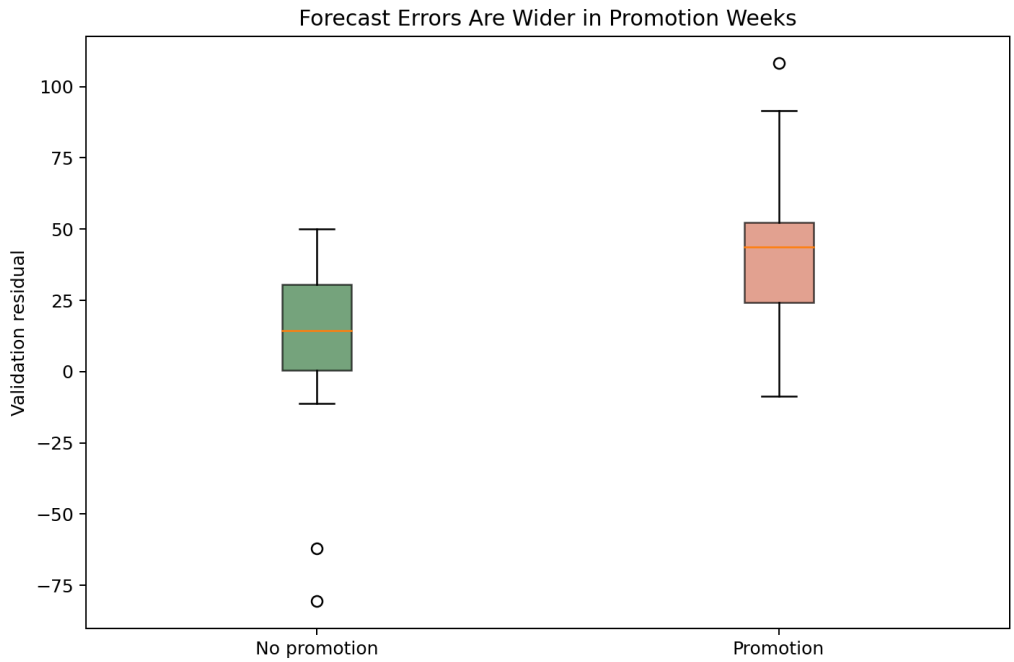

Figure 4: Example 5.4 notebook output: forecast errors widen sharply in high-VIX stressed-market weeks.

Decision interpretation

Definition: Decision threshold

A decision threshold is the business cutoff that matters operationally, such as a drawdown limit, a value-at-risk ceiling, or a minimum return required for action.

Forecast uncertainty becomes valuable only when it is linked to a decision. If the lower bound of the return interval breaches a portfolio drawdown limit, the risk manager may need to reduce exposure even when the point forecast looks benign. This is why Lecture 5 matters for practice: the forecast is not the final product, the decision is. A well-communicated interval helps managers think in terms of tail risk rather than false certainty.

Example 5.6. *In the notebook, a loss limit of -1.5% is defined as the exposure threshold. For each day in the test period, the model checks whether the lower bound of the 80% prediction interval falls below this limit. On flag days, a "reduce exposure" signal is issued even if the point forecast is near zero. This converts the statistical interval into a binary risk management rule.*

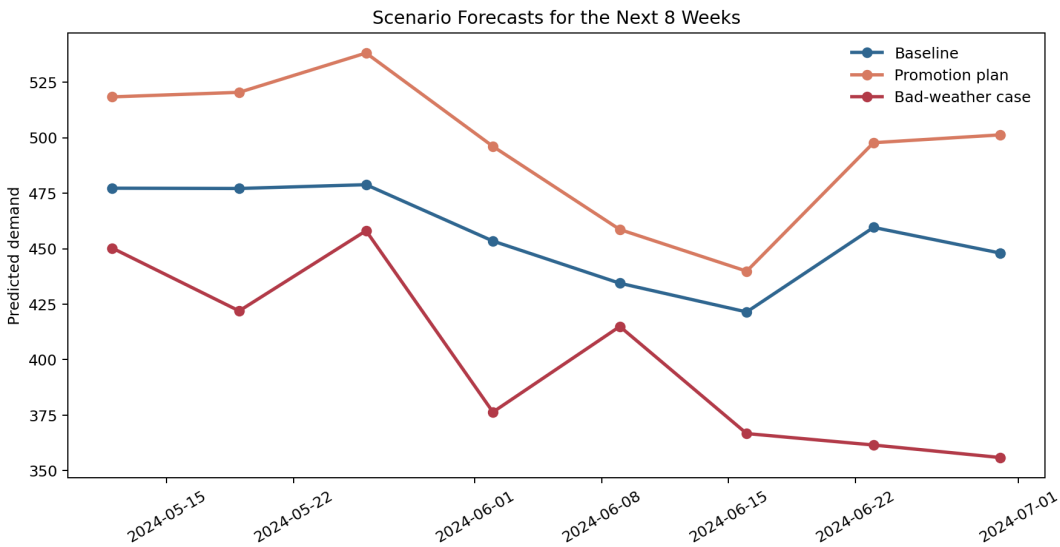

Figure 5: Example 5.5 notebook output: bull, neutral, and bear scenarios for S&P 500 cumulative returns.

Asymmetric costs of forecast errors

Definition: Asymmetric cost

Asymmetric cost means that over-forecasting and under-forecasting carry different penalties in a given business context.

Standard metrics like MAE and RMSE treat positive and negative errors symmetrically. But in most operational settings the two types of error are not equally bad.

- A portfolio manager who over-forecasts returns holds too much exposure and suffers a larger-than-expected drawdown.
- The same manager who under-forecasts returns reduces exposure unnecessarily and misses a rally.
- In risk-managed settings, the drawdown penalty is typically higher than the opportunity cost of a missed gain.

Asymmetric loss framework. Define forecast error $e_{t+1} = y_{t+1} - \hat{y}_{t+1|t}$. Assign different per-unit costs to over- and under-forecasting:

$$\text{Cost}(e) = \begin{cases} c_{\text{over}} \cdot |e| & \text{if } e < 0 \quad \text{(over-forecast: reality was lower than predicted)} \\ c_{\text{under}} \cdot |e| & \text{if } e > 0 \quad \text{(under-forecast: reality was higher than predicted)} \end{cases}$$

where c_{over} and c_{under} are cost coefficients set by business context.

If $c_{over} > c_{under}$, the firm prefers caution: better to reduce exposure early than to face an unexpected

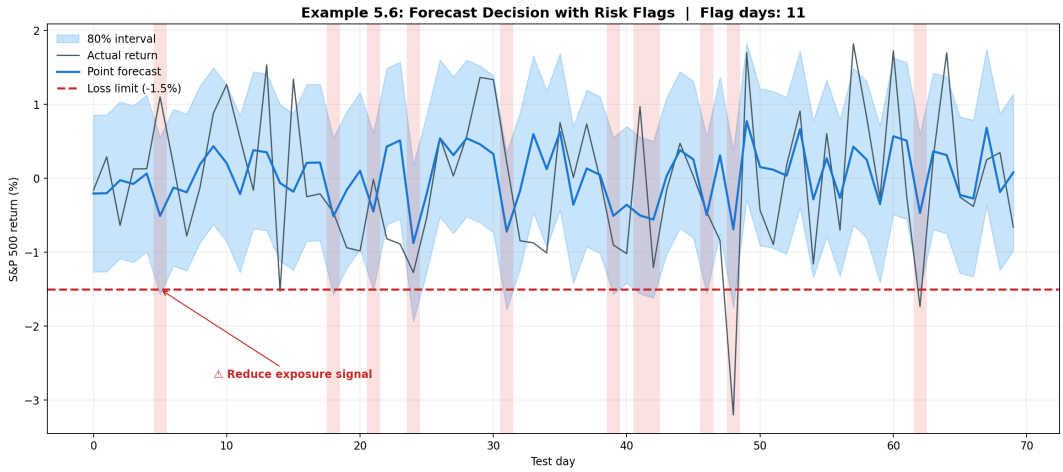

Figure 6: Example 5.6 notebook output: daily return interval with "reduce exposure" risk flags when the lower bound breaches the loss limit.

drawdown. This is the common setting in equity risk management, where downside losses tend to have greater operational and reputational cost than missed upside gains.

Practical implication. When business stakes are asymmetric, the optimal action is not to minimize symmetric RMSE but to shift the decision threshold:

1. Identify which direction of error is more expensive.
2. Use the prediction interval to locate the relevant percentile, not just the median.
3. Set the exposure or position limit to that percentile, not the center forecast.

Example 5.7. *In the notebook, an asymmetric cost framework is applied to the S&P 500 return model with $c_{over} = 3$ and $c_{under} = 1$, reflecting that a portfolio manager penalizes over-forecasting (holding excess exposure into a down move) three times more heavily than under-forecasting (reducing exposure and missing a rally). The optimal forecast percentile shifts from the median (50th) to the 75th percentile of the residual distribution, and the notebook compares total asymmetric cost under the symmetric median forecast versus the optimally shifted conservative forecast.*

Translating forecasts into actionable recommendations

Definition: Forecast-to-decision translation

Forecast-to-decision translation means converting a statistical output into a specific operational action, with explicit reasoning about uncertainty and cost.

A forecast is an input to a decision, not the decision itself. The final step of a forecasting workflow is: *given this forecast and its uncertainty, what should we do?*

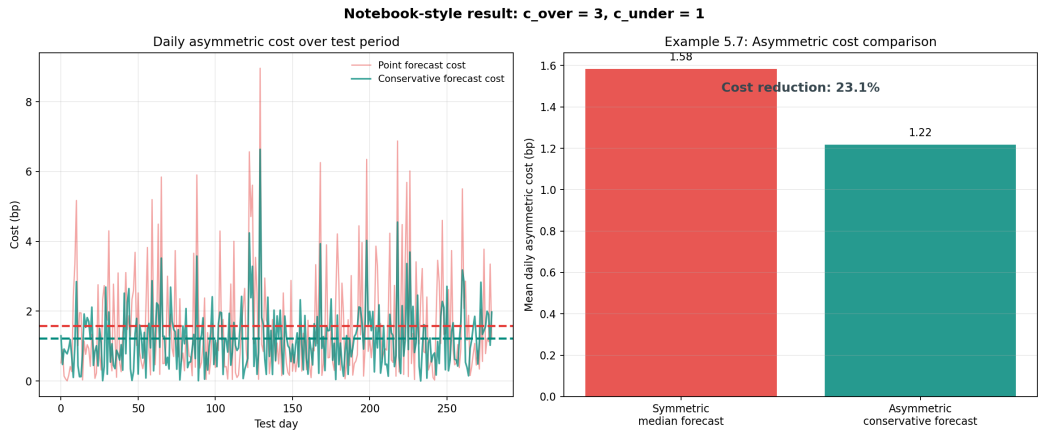

Figure 7: Example 5.7 notebook output: asymmetric cost comparison between symmetric-median and optimally shifted conservative forecasts.

Three-step translation template.

1. **State the forecast.** "Our model predicts next-day S&P 500 return of $+0.2\%$, with an 80% interval of -0.9% to $+1.3\%$."

2. **State the relevant threshold or constraint.** "Our risk limit is a -1.5% daily drawdown. The lower bound of -0.9% approaches but does not cross the limit—however, the 95% lower bound does."

3. **State the recommendation.** "We recommend a partial reduction in overnight exposure: keep core position but trim tail risk by reducing leverage to stay within drawdown limits even under the 95th-percentile downside."

The recommendation bridges statistics and risk operations. Without step 3, the forecast interval is a number with no action value. A risk manager needs a clear rule: when does the interval trigger a position change?

Interview-ready phrasing. Students should be able to say:

- *"The forecast suggests a mild positive return. But the 80% interval extends well into negative territory, which crosses our drawdown limit. My recommendation is to reduce exposure today."*

- *"The asymmetric cost here means I would rather under-invest than face a large drawdown. So I am using the 75th percentile of the conservative forecast, not the median."*

- *"Even though the point forecast is near zero, the lower tail of the interval breaches the risk threshold. That is why I flag this day for a reduce-exposure decision."*

Interview Questions This Lecture Prepares You To Answer

- **"So what would you recommend based on this forecast?"**
  "The point forecast is $+0.2\%$, but the 80% interval extends down to -0.9%. Given that over-forecasting in this risk context means holding excess exposure into a potential drawdown—which is more costly than missing a small rally—I would recommend reducing overnight exposure to stay inside the drawdown limit even under the lower tail."

- **"Which type of error is more costly here?"**
  "In this equity risk management context, over-forecasting the return (holding too much when the market falls) leads to drawdowns that exceed risk limits and damage portfolio mandates. Under-forecasting (holding too little when the market rallies) is costly but recoverable. So the model should be operated with a conservative bias: planning to the 75th percentile of the conditional return distribution rather than the median."

- **"How does this forecast change the decision?"**
  "Without the interval, the manager uses only the point forecast and may appear to have acceptable risk on every day. With the interval, we can identify specific days where the lower tail crosses the drawdown limit and trigger an exposure-reduction signal—even when the median forecast looks benign. That converts a passive statistical output into a proactive, rule-based risk management system."